🎓 MSIS-2407

DBMS - Design, Development & Administration

- 2 Units · 5 Weeks · Winter 2027 (weeks 6–10)
- Tuesday & Thursday, 7:35–9:10 pm — 10 sessions of 95 minutes
- Instructor: Dr. Mahmoud Parsian

📜 Catalog Description

MSIS-2407: Official course catalog description →

This course presents technical and managerial approaches to the analysis, design, and management of business data, databases, and database management systems.

The topics include structured and unstructured data management, a comparison of relational and object-oriented databases, relational database conceptual and logical design, and database implementation and administration.

🔗 Relationship to MSIS-2402

In **MSIS-2402** you learned:

- SQL querying
- Joins and aggregation
- Analytical SQL
- DuckDB analytics

In **MSIS-2407** you will learn:

How databases are designed, built, and managed.

🌍 Course Vision

Transform students from:

❌ SQL users



✅ Data system designers & decision-makers

🎯 Learning Outcomes

Students will be able to:

- Distinguish structured, semi-structured and unstructured data
- Design conceptual models with ER and Enhanced ER diagrams
- Convert them into logical relational schemas and normalize (1NF–BCNF, 4NF)
- Implement databases using SQL, with indexes and transactions
- Compare relational, object-oriented and document models
- Design data warehouses and generate business insights from them
- Secure, back up and govern a database

🧠 Learning Progression

Model

→ Normalize

→ Build

→ Analyze

→ Manage

📊 Course Structure

Week	Theme	Skill
1	Data Management and Conceptual Modeling	Structural thinking
2	Relational Design and Normalization	Dependency reasoning
3	Database Implementation	System building
4	Analytical and Non-Relational Data Systems	Model selection
5	Database Administration and Management	Decision-making

10 class sessions — two per week.

🗓️ How Each Session Works

- Lecture and live demonstration
- The session's **lab**, started in the last ~40 minutes of class
- Submitted by **11:00 pm** the same day

Every session has two Marimo labs — **a** (core) and **b** (a step more challenging) — each with built-in self-checks so you know when a task is right. Labs are published in the course repository's `LABS/` folder during class.

🧭 Catalog Alignment

Each topic named in the official course description maps to a week:

Catalog topic	Week
Structured and unstructured data management	1, 4
Relational database conceptual and logical design	1–2
Database implementation and administration	3, 5
Comparison of relational and object-oriented databases	4

Week 4 also covers data warehousing for analytical workloads — not named in the catalog, but central to MSIS/MSBA work.

🟦 Week 1 — Data Management and Conceptual Modeling

- Data management in organizations
- Structured, semi-structured & unstructured data (basics in Session 1; more in Week 4)
- Conceptual modeling: entities, attributes, relationships
- ER diagrams; cardinality and participation
- Enhanced ER (EER): superclass / subclass

🧪 Labs — Session 1: entities, keys, a first table split · Session 2: business rules → ER, then Enhanced ER

🟦 Week 2 — Relational Design and Normalization

- Mapping ER → relational schema
- Primary keys, foreign keys, integrity constraints
- Functional dependencies and data anomalies
- 1NF, 2NF, 3NF, BCNF (and 4NF); when to denormalize

🧪 Labs — Session 3: 1NF → 3NF, then BCNF · Session 4: keys and constraints, then 4NF

🟦 Week 3 — Database Implementation

- CREATE TABLE, data types, PK / FK constraints — in DuckDB, then MySQL
- Data loading and validation
- Indexing and query performance
- Transactions, ACID and concurrency

🧪 Labs — Session 5: typed DDL and loading messy data · Session 6: transactions and concurrency

🟦 Week 4 — Analytical and Non-Relational Data Systems

- OLTP vs OLAP; data warehouses
- Star schema: fact and dimension tables
- Grain, surrogate keys, SCD Type 2
- Relational vs object-oriented and document models (STRUCT, LIST, JSON)
- Semi-structured data in depth: schema drift, shredding JSON into tables
- Data lakes, lakehouse, medallion architecture

🧪 Labs — Session 7: OLTP → star schema, SCD Type 2 · Session 8: querying a warehouse; relational vs document models

🟦 Week 5 — Database Administration and Management

- The DBA role: performance and monitoring
- Security and access control (views, PII, SQL injection)
- Backup, recovery (RPO / RTO) and capacity
- Data quality and governance

🧪 Labs — Session 9: security, access, masking, indexes, auditing · Session 10: backup, recovery, data quality

📦 Assessments

Component	Points
🧪 10 Labs — one per class session, 70 points each	700
✏️ Final Exam — 2 hours, separate exam slot after Session 10	300
Total	**1000**

Each lab is started in class and submitted by 11:00 pm the same day.
Details: `course_information/ASSIGNMENTS_and_GRADING.md`

🧪 Tools & Technologies

- SQL
- **DuckDB** — primary engine for all labs and notebooks
- **MySQL** — transactional-systems contrast (Week 3)
- **Marimo** — reactive Python notebooks for the labs
- ER modeling tools (draw.io, Lucidchart)

💼 Teaching Philosophy

- Learn by doing
- Emphasize real-world problems
- Focus on business impact

🧠 Cognitive Levels (Bloom's Taxonomy)

Level	Application
Understand	Data types and data modeling
Apply	SQL implementation
Analyze	Normalization and dependency reasoning
Evaluate	Data-model and performance trade-offs
Create	Schemas, star schemas, and role views built in the labs

🎯 Final Outcome

After completing this course you will be able to:

- Design scalable databases
- Implement and administer working systems
- Build analytical data models and deliver actionable insights
- Understand how enterprise data platforms operate

🚀 Career Relevance

Prepares students for roles:

- Data Analyst
- Data Engineer (entry-level)
- Business Intelligence Analyst

📂 Where to Find Things

All in the course repository:

Folder	What's there
`software_installation/`	Setup guide — do Steps 0–3 **before the first class**
`course_information/`	Grading, office hours, laptop rule, required software
`week1_…` – `week5_…`	Lecture notes, slides, and demo notebooks
`LABS/`	Each session's labs, published during class

🙌 End of Syllabus